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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

4274470357

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **October/November 2012**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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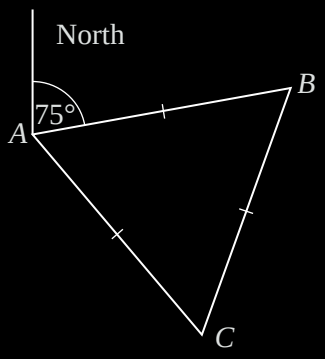
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Answer .....[1]

(b) Find the bearing of *C* from *B*.

Answer .....[1]

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Number of Fives	0	1	2	3	4
Frequency	1	2	3	5	1

For this distribution of Fives,

(a) write down the mode,

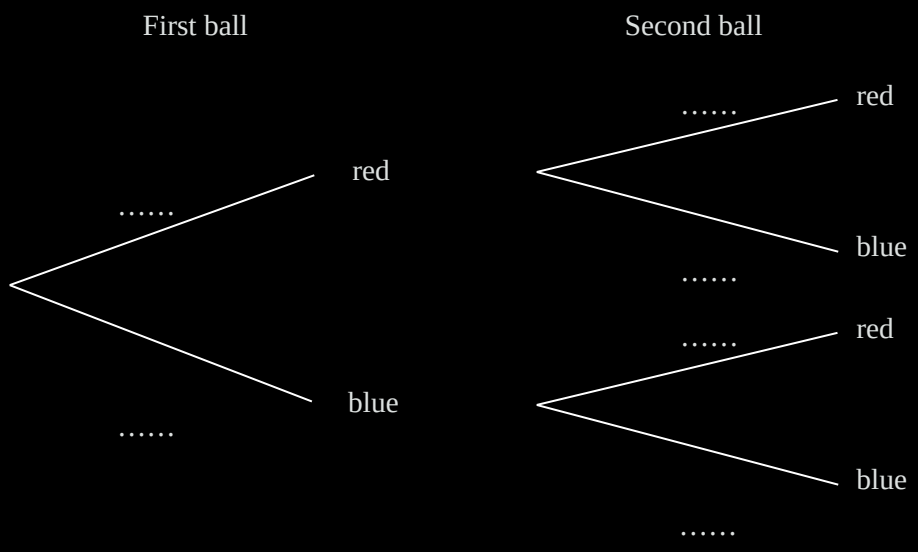
Answer ..... [1]

(b) find the median.

Answer ..... [1]

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12 A bag contains 1 red and 3 blue balls.
Two balls are taken from the bag, at random, without replacement.
The tree diagram that represents all the outcomes is shown below.



(a) Write the appropriate probability on each branch. [2]

(b) Find the probability that the second ball taken is red.

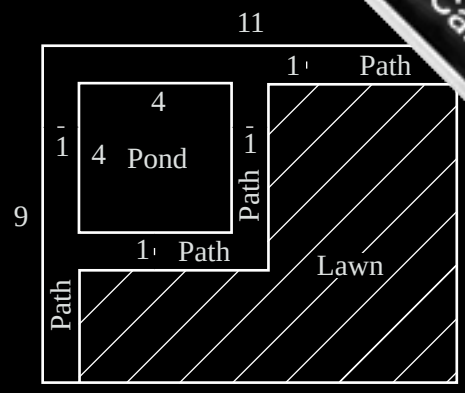
Answer ..... [1]

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garden, measuring 11 m by 9 m.
All the angles are right angles.
The pond is a square of side 4 m.
The paths are 1 m wide.
The remainder of the garden is a lawn.



(a) Find the perimeter of the lawn.

Answer ..... m [1]

(b) Find the total area of the paths.

Answer ..... m^2 [1]

(c) The paths are paved with square tiles of side 50 cm.

How many tiles are used?

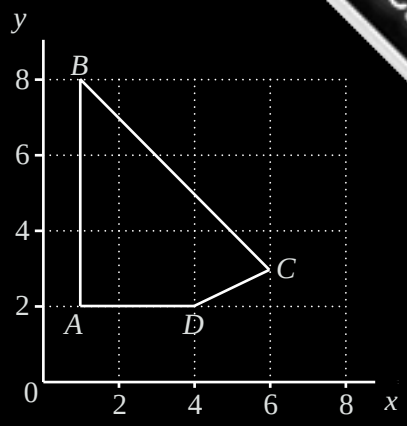
Answer .....[1]

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$x = 1$, $y = 2$, $2y = x$ and $x + y = 9$.

The region **inside** the quadrilateral is defined by four inequalities.
Two of these are $y > 2$ and $2y > x$.



(a) Write down the other two inequalities.

Answer .....

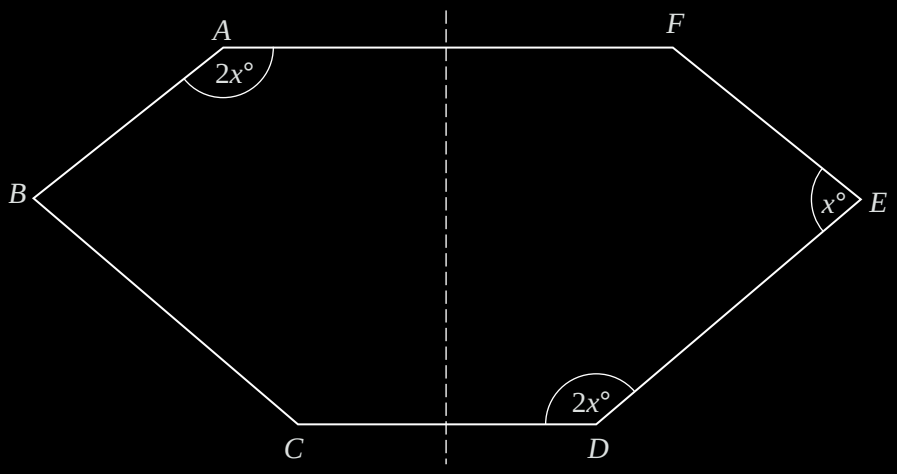
..... [2]

(b) How many points, with **integer** coordinates, lie **inside** the quadrilateral *ABCD*?

Answer ..... [1]

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In the diagram, the dashed line is a line of symmetry.
$B\hat{A}F = 2x°$, $F\hat{E}D = x°$ and $C\hat{D}E = 2x°$.

Find the value of x.

Answer $x =$ ..... [3]

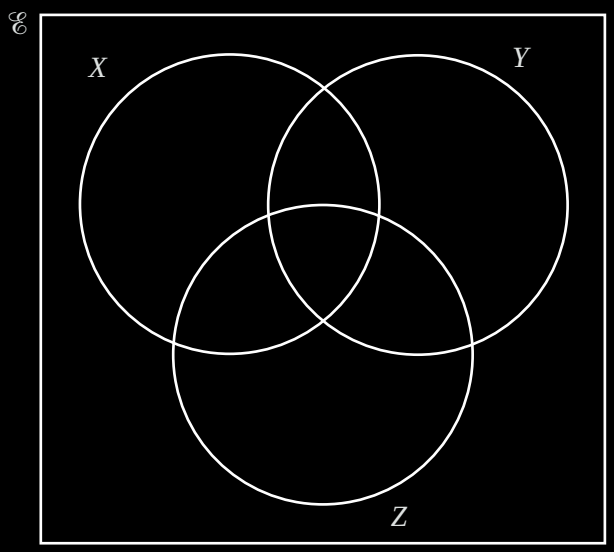
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On the Venn diagram, write each of the letters p, q, and r in its appropriate subset, given that

$p \in X \cap Y \cap Z$,

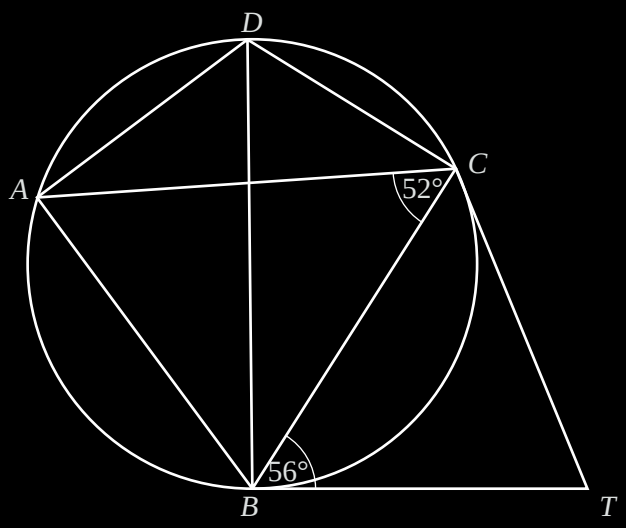
$q \in X' \cap Y' \cap Z'$,

$r \in (X \cup Y)' \cap Z$.



[3]

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In the diagram, the points *A*, *B*, *C* and *D* lie on the circle.
BD is a diameter.
The tangents from *T* touch the circle at *B* and *C*.
$A\hat{C}B = 52°$ and $T\hat{B}C = 56°$.

Find

(a) $B\hat{T}C$,

Answer $B\hat{T}C =$ ..... [1]

(b) $A\hat{D}B$,

Answer $A\hat{D}B =$ ..... [1]

(c) $B\hat{D}C$,

Answer $B\hat{D}C =$ ..... [1]

(d) $A\hat{B}C$.

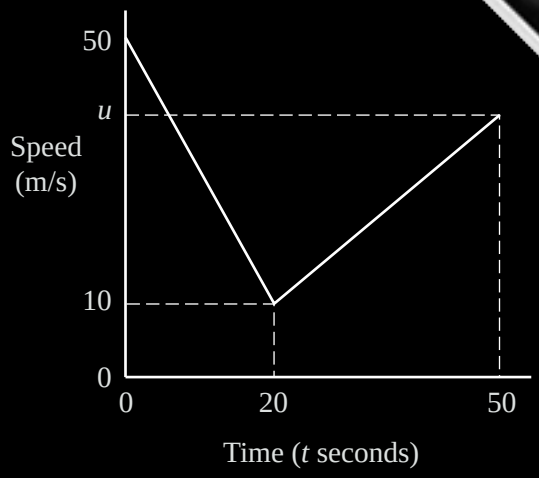
Answer $A\hat{B}C =$ ..... [1]

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part of a train's journey.

The train slows down uniformly from a speed of 50 m/s to a speed of 10 m/s in a time of 20 seconds.

During the next 30 seconds, it accelerates uniformly to a speed of u metres/second.



(a) Calculate the retardation from $t = 0$ to $t = 20$.

Answer ..... m/s^2 [1]

(b) Calculate the speed of the train when $t = 15$.

Answer ..... m/s [1]

(c) Calculate the distance travelled by the train from $t = 0$ to $t = 20$.

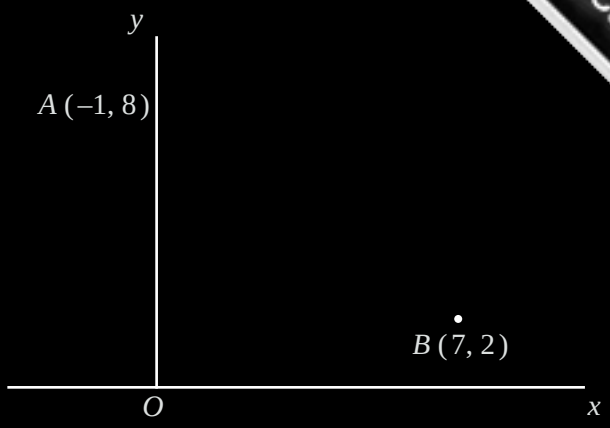
Answer ..... m [1]

(d) The size of the acceleration is half the size of the retardation.
Find the value of u.

Answer $u =$ ..... [1]

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A (–1, 8) and B (7, 2).



(a) Find the coordinates of the midpoint of AB.

Answer (..... , ...) [1]

(b) $\overrightarrow{BC} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$

(i) Find the coordinates of C.

Answer (..... , ...) [1]

(ii) Given that $\left|\overrightarrow{AB} + \overrightarrow{BC}\right| = \sqrt{k}$, find k.

Answer $k =$ .....[2]

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Row number	1	2	3	4		*n*
Square of the middle number in the row	4	25	64			*x*
Product of the first and the last number in the row	3	24	63			*y*

(i) Complete the column for Row number 4. [1]

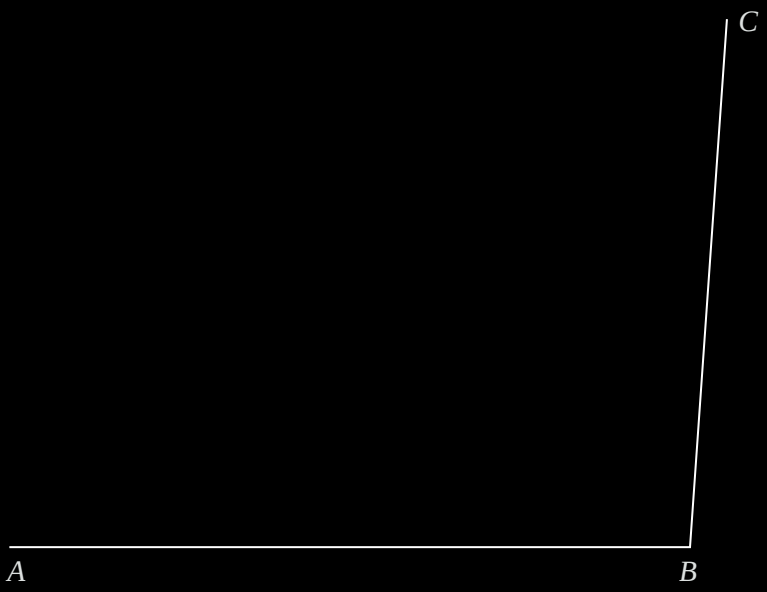
(ii) Find an expression, in terms of *n*, for *y*.

Answer ..... [1]

(iii) Show that $x - y$ is always equal to 1.

[2]

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